Referee Data Memo: Fiscal-LPT Sectoral C2-SUT Calibration Database

Codex review memo

May 14, 2026

Contents

1 Purpose and Scope

This memo explains the database currently used to construct the Fiscal-LPT production-wedge calibration in the MCMS Julia runtime. It is written for a referee who needs to understand what the database contains, where each entry comes from, how the entries are transformed into model objects, and which measurement faults remain.

The active production-wedge runtime is now the sectoral C2-SUT layer. It lives in:

```
C:\CustomTools\MCMS_gvt\MCMS-private\input_data\fiscal_rules_sectoral_c2_sut
```

The active sectoral builder is:

```
C:\CustomTools\MCMS_gvt\MCMS-private\scripts\build_fiscal_rule_dataset_sectoral_c2_sut.py
```

The Julia runtime imports the generated file:

```
C:\CustomTools\MCMS_gvt\MCMS-private\input_data\fiscal_rules_sectoral_c2_sut\sectoral_c2_sut_runtime.jl
```

from:

```
C:\CustomTools\MCMS_gvt\MCMS-private\src\calibration.jl
```

This memo deliberately focuses on data construction and calibration provenance. Model-solution diagnostics such as BK conditions are out of scope here, per the current instruction. The memo records the data limitations that would remain even if the model-solution issue were ignored.

2 Executive Summary

The current runtime production-wedge database is a sectoral calibration built from BEA Supply-Use tables. It replaces the earlier neutral sectoral allocation of aggregate Quarterly C2 coefficients with 44 sector-specific production-wedge rules.

The core design is:

1. Estimate U.S. model-sector production-wedge rules from annual BEA Supply-Use rows.
2. Remove the BEA SUT row `Taxes on products and imports`, which includes import duties, so the measured wedge is closer to a non-product production tax/subsidy margin.
3. Map BEA summary industries into the 44 MCMS model sectors.
4. Transfer the U.S. sectoral rules to EA, China, and ROW using the existing C2 country scaling inputs. EA uses a direct Eurostat D29 scaling layer; China and ROW remain broad-fiscal fallback transfer blocks.
5. Keep unrestricted empirical debt-feedback columns in the CSV audit files, but use a runtime sign closure that clips negative production-wedge debt feedbacks to zero.
6. Keep the steady-state production wedge as the model-implied markup-offsetting object `Tbar_Y = 1/M - 1`, not as observed fiscal data.

The database is transparent and reproducible, but it is not a structural Leeper-Plante-Traum estimation. It is an annual sectoral reduced-form calibration layer that feeds the MCMS fiscal block. The earlier aggregate Quarterly C2 files remain useful as supporting evidence and provenance, but they are no longer the source of the active production-wedge matrices in `calibration.jl`.

3 Active Runtime Objects

The model rule for the production wedge is:

```
tau_Y[k,i,t] =
    rho_Y[k,i] * tau_Y[k,i,t-1]
  + (1 - rho_Y[k,i]) * (
        Phi_Y_y[k,i] * output_gap[k,t]
      + Phi_Y_b[k,i] * bgov_dev[k,t-1]
    )
  + Sigma_Y[k,i] * eps_Y[k,i,t]
```

The active runtime matrices are generated in `sectoral_c2_sut_runtime.jl` and imported by `calibration.jl`. Country order is:

```
EA, China, ROW, USA
```

The model now assigns the sectoral matrices directly:

```
sectoral_c2_sut = sectoral_c2_sut_runtime_matrices()
Rho_tau_Y       = sectoral_c2_sut.Rho_tau_Y
Phi_tau_Y_y     = sectoral_c2_sut.Phi_tau_Y_y
Phi_tau_Y_b     = sectoral_c2_sut.Phi_tau_Y_b
Sigma_tau_Y     = sectoral_c2_sut.Sigma_tau_Y
```

The active matrices have shape 4-by-44. The following table summarizes the runtime range by country:

Country	Sectors	`Rho` min	`Rho` max	`Phi_y` min	`Phi_y` max	`Phi_b` min	`Phi_b` max	`Sigma` min	`Sigma` max
EA	44	0.335170	0.673505	-0.033992	0.000813	0.000000	0.094691	0.000234	0.013500
China	44	0.228933	0.567268	-0.272646	0.006463	0.000000	0.138534	0.000945	0.054585
ROW	44	0.292710	0.631045	-0.020092	0.000479	0.000000	0.169720	0.000541	0.031249
USA	44	0.161373	0.838042	-0.145942	0.007139	0.000000	0.025032	0.000688	0.039754

The generated coefficient file keeps both unrestricted empirical debt-feedback columns and runtime debt-feedback columns. The runtime uses a closure rule named:

```
clip_negative_phi_b_to_zero
```

This distinction matters. Negative unrestricted sectoral debt responses are not deleted from the audit file, but they are not passed into the Julia runtime. The rationale is economic and computational rather than empirical: with the model sign convention, a negative production-wedge debt feedback means that higher inherited debt lowers production taxes or raises subsidies. The runtime therefore prevents the production-wedge instrument from becoming a debt-destabilizing margin and leaves debt stabilization to other instruments or non-negative production-wedge debt responses.

4 Source Inventory

4.1 U.S. Quarterly Sources

All U.S. active production-wedge source series are pulled from FRED, which mirrors BEA National Income and Product Account or industry-account series. FRED is used because it gives stable CSV endpoints and public source metadata.

Internal column	FRED series	Description	Frequency	Unit	Origin website
`us_topi_less_subsidies_saar`	`W254RC1Q027SBEA`	Taxes on production and imports less subsidies	Quarterly	Billions of dollars, seasonally adjusted annual rate	https://fred.stlouisfed.org/series/W254RC1Q027SBEA
`us_federal_excise_taxes_saar`	`B234RC1Q027SBEA`	Federal excise taxes	Quarterly	Billions of dollars, seasonally adjusted annual rate	https://fred.stlouisfed.org/series/B234RC1Q027SBEA
`us_customs_duties_saar`	`B235RC1Q027SBEA`	Federal customs duties	Quarterly	Billions of dollars, seasonally adjusted annual rate	https://fred.stlouisfed.org/series/B235RC1Q027SBEA
`us_state_local_sales_taxes_saar`	`B248RC1Q027SBEA`	State and local sales taxes	Quarterly	Billions of dollars, seasonally adjusted annual rate	https://fred.stlouisfed.org/series/B248RC1Q027SBEA

Internal column	FRED series	Description	Frequency	Unit	Origin website
`us_gross_output_billions_saar`	GOAI	Gross output by industry, all industries	Quarterly	Billions of dollars, seasonally adjusted annual rate	https://fred.stlouisfed.org/series/GOAI
`us_gdp_billions_saar`	GDP	Nominal GDP	Quarterly	Billions of dollars, seasonally adjusted annual rate	https://fred.stlouisfed.org/series/GDP
`us_real_gdp_billions_2017_saar`	GDPC1	Real GDP	Quarterly	Billions of chained 2017 dollars, seasonally adjusted annual rate	https://fred.stlouisfed.org/series/GDPC1
`us_federal_debt_pct_gdp`	GFDEGDQ188S	Federal debt as percent of GDP	Quarterly	Percent of GDP	https://fred.stlouisfed.org/series/GFDEGDQ188S

FRED's public metadata for `W254RC1Q027SBEA` identifies BEA as the source, reports quarterly frequency, and reports the unit as billions of dollars at a seasonally adjusted annual rate. The same FRED metadata structure is used for the other U.S. series.

4.2 Euro Area Quarterly Sources

The euro-area source block is pulled from Eurostat API endpoints.

Internal object	Eurostat dataset / item	Description	Frequency	Unit	Origin website
`D29REC`	`gov_10q_ggnfa`	Other taxes on production, revenue	Quarterly	Percent of GDP	https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/gov_10q_ggnfa?format=JSON&lang=en&freq=Q&unit=PC_GDP&sector=S13&geo=EA20
`D39PAY`	`gov_10q_ggnfa`	Other subsidies on production, expenditure	Quarterly	Percent of GDP	Same endpoint

Internal object	Eurostat dataset / item	Description	Frequency	Unit	Origin website
`GD`	`gov_10q_ggdebt`	General-government gross debt	Quarterly	Percent of GDP	https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/gov_10q_ggdebt?format=JSON&lang=en&freq=Q&unit=PC_GDP&sector=S13&geo=EA20&na_item=GD
`B1GQ`	`namq_10_gdp`	GDP at market prices, chain-linked volumes	Quarterly	Chain-linked volume, million euro	https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/namq_10_gdp?format=JSON&lang=en&freq=Q&unit=CLV15_MEUR&s_adj=SCA&na_item=B1GQ&geo=EA20

The generated metadata records:

```
gov_10q_ggnfa updated: 2026-04-22T11:00:00+0200
gov_10q_ggdebt updated: 2026-04-22T11:00:00+0200
namq_10_gdp updated: 2026-05-13T11:00:00+0200
government-finance seasonal adjustment used: NSA
real GDP seasonal adjustment used: SCA
```

Eurostat's public metadata describes `gov_10q_ggdebt` as quarterly general-government debt measured at the end of each quarter, compiled under ESA 2010, and available in percent of GDP. Eurostat metadata for `gov_10q_ggnfa` describes quarterly non-financial accounts for general government. Eurostat metadata for `namq_10_gdp` describes quarterly national accounts, including chain-linked volume measures.

4.3 Active Sectoral SUT Sources

The active U.S. sectoral source layer lives in:

```
MCMS-private\input_data\fiscal_rules_sectoral_c2_sut
```

The source is BEA Supply-Use data:

```
https://apps.bea.gov/industry/release/zip/SUPPLY-USE.zip
```

The generated metadata records:

```
content_type: application/x-zip-compressed
content_length: 4791455
last_modified: Thu, 25 Sep 2025 12:30:06 GMT
```

The sectoral builder reads `Use_Summary.xlsx` inside that zip file. For each BEA summary industry and year, it constructs:

```
C2_SUT_wedge =
    Other taxes on production
  - Other subsidies on production
  - Subsidies on products
```

and divides by total industry output at basic prices. It excludes the BEA SUT row:

```
Taxes on products and imports
```

This SUT layer is now the active dynamic production-wedge runtime. It is not, however, a full international sectoral fiscal database. It is annual, the underlying sector-specific estimates are U.S.-only, and it depends on a mapping from BEA summary industries into the model's 44 sectors. EA, China, and ROW are constructed by scaling these U.S. sectoral rules with C2 country-level scaling inputs.

5 Supporting Aggregate U.S. Quarterly C2 Wedge

The earlier aggregate U.S. Quarterly C2 layer remains useful as supporting country-level evidence. Its numerator is:

```
US_C2_numerator =
    taxes_on_production_and_imports_less_subsidies
  - federal_excise_taxes
  - customs_duties
  - state_local_general_sales_taxes
```

In source names:

```
US_C2_numerator =
    W254RC1Q027SBEA
  - B234RC1Q027SBEA
  - B235RC1Q027SBEA
  - B248RC1Q027SBEA
```

The U.S. production-wedge level is:

```
US_C2_wedge = US_C2_numerator / GOAI
```

This construction removes customs duties, so tariff revenue is not loaded into the production-wedge rule. It also removes federal excise and state/local sales taxes, which are product-tax or consumption-tax-like components.

The main remaining U.S. source limitation is that the quarterly public series does not expose every product-tax subcomponent available in the richer annual NIPA decomposition. The quarterly construction is therefore a major observable product-tax exclusion, not a perfect annual-style product-tax exclusion.

6 Supporting Aggregate Euro-Area Quarterly C2 Wedge

The supporting aggregate euro-area quarterly object is:

```
EA_C2_wedge = D29REC - D39PAY
```

where:

```
D29REC = other taxes on production, revenue
D39PAY = other subsidies on production, expenditure
```

Eurostat reports both as percent of GDP in the API call used by the builder. The interpretation is a non-product firm-side production-tax/subsidy wedge at quarterly frequency.

The strongest feature of this construction is concept fit: D29 and D39 are much closer to non-product production taxes and subsidies than a broad taxes-on- production-and-imports measure. The main limitation is denominator mismatch: the EA object is a GDP share, while the U.S. object is a gross-output share. This matters because the model production wedge is naturally closer to a firm-side cost wedge than to a GDP revenue ratio. The current builder accepts this mismatch in order to use direct quarterly public data.

7 International Transfer Construction

EA, China, and ROW do not have integrated direct sectoral production-tax and production-subsidy rules comparable to the U.S. BEA SUT panel. They are constructed by scaling the U.S. sectoral C2-SUT rules with existing C2 country-level scaling inputs.

For each non-U.S. country block:

1. Start from the 44 U.S. sectoral C2-SUT production-wedge rules.
2. Use the existing C2 country fiscal-responsiveness scaling inputs from:

```
MCMS-private\input_data\fiscal_rules_calibration_c2\calibration_c2_country_scaling_inputs.csv
```

3. Apply the country-specific tax-margin, output-feedback, debt-feedback, shock-scale, and persistence proxies in that file.
4. For EA, use the direct Eurostat D29 scaling layer recorded in the C2 scaling file.
5. For China and ROW, keep the broad-fiscal absolute fallback treatment and inherit the U.S. sectoral sign pattern where direct production-tax data are unavailable.

EA, China, and ROW should therefore be described as calibrated transfer blocks for sectoral fiscal dynamics, not as direct country-sector production-tax estimates.

8 Harmonisation Steps

8.1 Country Order

All runtime matrices are written in the MCMS runtime country order:

```
EA, China, ROW, USA
```

This order is repeated in the generated sectoral CSVs, the generated `sectoral_c2_sut_runtime.jl` file, and `src/calibration.jl`.

8.2 Frequency

The active sectoral C2-SUT calibration is annual because the BEA SUT source is annual. This means the latest update reintroduces a frequency limitation: the active sectoral rules have better sectoral tax-base measurement but weaker model-frequency alignment than the aggregate Quarterly C2 layer.

The aggregate Quarterly C2 layer remains available as supporting country-level evidence, but `calibration.jl` now uses the sectoral C2-SUT matrices for `Rho_tau_Y`, `Phi_tau_Y_y`, `Phi_tau_Y_b`, and `Sigma_tau_Y`.

8.3 Model Deviations

For the aggregate quarterly EA and USA evidence, observed wedge series are converted to model deviations around latest-20-quarter means:

```
tau_Y_model_dev =
    log(1 + observed_wedge_t)
  - log(1 + latest20_average_wedge)
```

Public debt is converted into an additive deviation:

```
bgov_dev_t =
    debt_gdp_t
  - latest20_average_debt_gdp
```

For the active sectoral C2-SUT layer, each U.S. sectoral annual wedge is converted relative to a latest-five-year sectoral benchmark:

```
tau_Y_us_sector_c2_sut_model_dev =
    log(1 + c2_sut_non_product_share_output_t)
  - log(1 + sector_tbar_c2_sut_latest5)
```

In both layers, `tau_Y` is a log gross-wedge deviation, while public debt is an additive deviation.

8.4 Output Gap

The output gap is a reduced-form statistical gap. For each directly observed country block, the builder:

1. takes real GDP,
2. logs it,
3. fits a linear trend in quarterly time,
4. uses the residual as `output_gap_log_linear`.

This is transparent and reproducible, but it is not a model-consistent output gap from the DSGE solution.

8.5 Regression Form

The quarterly rule is estimated without intercept:

```
tau_t =
    rho * tau_(t-1)
  + beta_y * output_gap_t
  + beta_b * bgov_dev_(t-1)
  + residual_t
```

The no-intercept specification is chosen because the dependent variable and debt variable are already centered around recent steady-state benchmarks. It also aligns the estimated object with a model rule that has no constant term in deviation form.

8.6 Model-Rule Conversion

The implemented model rule uses:

```
(1-rho) * Phi
```

inside the fiscal feedback term. Therefore the reduced-form beta coefficients are converted into model-rule coefficients as:

```
Phi_y = beta_y / (1-rho)
Phi_b = beta_b / (1-rho)
```

The generated files keep both unrestricted empirical beta/Phi columns and the runtime beta/Phi columns after the debt-feedback closure rule.

8.7 Runtime Debt-Feedback Closure

The unrestricted empirical sectoral estimates can imply negative production-wedge debt feedbacks. The generated runtime columns impose:

```
Phi_tau_Y_b = 0 when empirical Phi_tau_Y_b < 0
Beta_tau_Y_b = 0 when empirical Beta_tau_Y_b < 0
```

The closure is not an empirical estimate. It is a policy/modeling restriction used to prevent the production-wedge instrument from lowering taxes or raising subsidies in response to high debt.

9 Generated Files and Entries

The active sectoral C2-SUT folder contains these files:

File	Role
`metadata.json`	Source URLs, generation principle, package versions, generated-file manifest
`raw_bea_sut_summary_tax_rows.csv`	BEA SUT summary-industry tax rows and wedge components
`sector_mapping_model_to_bea_summary.csv`	Explicit mapping from 44 model sectors to BEA summary industries
`us_model_sector_c2_sut_wedges.csv`	Annual U.S. sectoral wedge levels by model sector
`us_model_sector_c2_sut_panel.csv`	U.S. sectoral panel with output gap, debt deviation, latest-five-year benchmark, and model deviations
`us_model_sector_c2_sut_rule_estimates.csv`	Direct U.S. sector-by-sector reduced-form rule estimates
`sectoral_c2_sut_scaled_tau_y_coefficients.csv`	176 country-sector empirical and runtime coefficients after country scaling
`Rho_tau_Y_sectoral_c2_sut_matrix.csv`	Active 4-by-44 persistence matrix
`Phi_tau_Y_y_sectoral_c2_sut_matrix.csv`	Active 4-by-44 output-feedback matrix
`Phi_tau_Y_b_sectoral_c2_sut_matrix.csv`	Active 4-by-44 runtime debt-feedback matrix

File	Role
`Sigma_tau_Y_sectoral_c2_sut_annual_matrix.csv`	Active 4-by-44 annual shock-scale matrix
`sectoral_c2_sut_runtime.jl`	Generated Julia runtime matrices imported by `src/calibration.jl`
`sectoral_c2_sut_audit.md`	Machine-generated audit note

The scaled country-sector coefficient file has the most important entries:

Column	Meaning
`model_country`	MCMS country block in runtime order
`model_sector_index`	MCMS sector index, 1 to 44
`model_sector`	MCMS sector code
`bea_summary_codes`	BEA summary industry code or codes mapped into the model sector
`mapping_note`	Manual mapping caveat where BEA and model classifications do not align cleanly
`Rho_tau_Y_sectoral_c2_sut`	Runtime persistence in the sectoral production-wedge rule
`Beta_tau_Y_y_empirical_sectoral_c2_sut`	Unrestricted reduced-form output-gap beta
`Beta_tau_Y_b_empirical_sectoral_c2_sut`	Unrestricted reduced-form debt beta
`Phi_tau_Y_y_empirical_sectoral_c2_sut`	Unrestricted output-feedback model-rule coefficient
`Phi_tau_Y_b_empirical_sectoral_c2_sut`	Unrestricted debt-feedback model-rule coefficient
`Beta_tau_Y_y_sectoral_c2_sut`	Runtime output-gap beta
`Beta_tau_Y_b_sectoral_c2_sut`	Runtime debt beta after closure
`Phi_tau_Y_y_sectoral_c2_sut`	Runtime output-feedback coefficient
`Phi_tau_Y_b_sectoral_c2_sut`	Runtime debt-feedback coefficient after closure
`Sigma_tau_Y_sectoral_c2_sut_annual`	Annual residual shock scale
`debt_feedback_closure`	Name of the rule mapping unrestricted debt feedback to runtime debt feedback
`tax_margin_scale`	Country scaling factor for the non-product tax margin
`fiscal_y_scale`	Country scaling factor for output-gap responsiveness
`fiscal_b_scale`	Country scaling factor for debt responsiveness
`fiscal_sigma_scale`	Country scaling factor for shock scale
`fiscal_rho_proxy`	Country persistence proxy used when transferring U.S. sectoral rules
`fiscal_proxy_type`	Whether scaling is direct or fallback
`source_status`	Direct or transfer-calibrated provenance statement

10 Supporting Aggregate Quarterly Estimates

The aggregate Quarterly C2 files still report direct country-level estimates. They are useful as a frequency-aligned diagnostic and as provenance for the C2 update, but they are not the active source of `Rho_tau_Y`, `Phi_tau_Y_y`, `Phi_tau_Y_b`, and `Sigma_tau_Y` after the sectoral update.

The direct U.S. aggregate estimate uses 83 quarterly observations from 2005Q2 to 2025Q4:

Object	Value
rho	0.688885
beta__y	0.003151
beta__b	-0.002875
Phi__y	0.010127

Object	Value
Phi_b	-0.009242
sigma	0.003484
R2, no intercept	0.622560

The direct EA aggregate estimate uses 95 quarterly observations from 2002-Q2 to 2025-Q4:

Object	Value
rho	0.063775
beta_y	0.055111
beta_b	-0.014042
Phi_y	0.058865
Phi_b	-0.014999
sigma	0.005259
R2, no intercept	0.209047

The low EA no-intercept R2 is not fatal for a calibration object, but it should be disclosed. It means the EA quarterly production-wedge movement is only weakly explained by lagged wedge, trend-output gap, and lagged debt deviation in this simple reduced-form rule.

11 Sectoral Calibration

The active runtime no longer repeats country aggregate rates across sectors. The U.S. row of each 4-by-44 runtime matrix is estimated sector by sector from the BEA SUT panel. The non-U.S. rows preserve this U.S. sectoral pattern but scale it with country-level C2 factors.

The U.S. sectoral estimation equation is:

```
tau_Y_sector[s,t] =
    rho_s * tau_Y_sector[s,t-1]
  + beta_y_s * output_gap[t]
  + beta_b_s * bgov_dev[t-1]
  + residual[s,t]
```

The model-rule conversion remains:

```
Phi_y_s = beta_y_s / (1-rho_s)
Phi_b_s = beta_b_s / (1-rho_s)
```

The panel covers 44 U.S. model sectors and 1997-2024 annual SUT observations. The rule estimates use 27 regression observations per U.S. sector after the lag. All 44 U.S. sectoral rules are estimated; none are missing.

Important mapping caveats remain:

1. Model sector `01T02` uses BEA `111CA`, so forestry is not separately identified.
2. Model sector `03` uses BEA `113FF`, which also contains forestry and related activities.
3. Chemicals and pharmaceuticals, model sectors `20` and `21`, both use BEA `325`.
4. Several services sectors combine multiple BEA summary industries because the model and BEA classifications are not one-to-one.

12 Relation to LPT

The current database is LPT-style only in the narrow sense that fiscal instruments respond to output and inherited debt in a partial-adjustment rule. It is not an LPT estimation.

The differences are:

1. LPT estimates fiscal rules inside a DSGE model.
2. The current database estimates reduced-form OLS rules outside the DSGE.
3. LPT is a structural posterior exercise; this is a data bridge plus calibration restriction.
4. The current database estimates only the production-wedge dynamic rule. Consumption-tax, labour-tax, and several steady-state fiscal objects remain benchmarked or model-implied.

The safe language is:

```
The fiscal block uses an LPT-style partial-adjustment debt-feedback structure.
The active production-wedge coefficients are annual sectoral reduced-form
calibration objects, not structural LPT posterior estimates.
```

13 Remaining Faults and Open Issues

13.1 EA, China, and ROW sectoral rules are not directly estimated

EA, China, and ROW are transfer calibrations at the sectoral level. The database does not contain direct sector-by-sector production-tax or production-subsidy rules for those blocks. Their coefficients are scaled from the U.S. sectoral C2-SUT rule using existing C2 responsiveness proxies.

13.2 ROW is not residual ROW

ROW should ideally be the world aggregate net of EA, China, and USA. The current supporting scaling infrastructure treats ROW as a world proxy in older data layers. This is not the same object as the residual MCMS rest of world.

13.3 Sectoral detail comes at annual frequency

The latest update improves sectoral measurement but gives up the quarterly frequency of the aggregate C2 layer. The active matrices are derived from annual BEA SUT observations. A referee could reasonably ask whether annual sectoral variation should be transformed or rescaled before entering a quarterly model.

13.4 U.S. and EA denominators differ in the supporting aggregate layer

The U.S. wedge is measured over gross output. The EA wedge is measured as a percent of GDP. This is a substantive harmonisation issue. Gross output is closer to a production cost base; GDP is closer to an aggregate revenue base. The difference is accepted because direct quarterly public EA gross-output denominators were not integrated.

13.5 U.S. sector mapping is explicit but imperfect

The sectoral SUT layer maps BEA summary industries into 44 model sectors. This is transparent in `sector_mapping_model_to_bea_summary.csv`, but it is not always one-to-one. The largest visible compromises are forestry/fishing, chemicals/pharmaceuticals, water and waste, transport support, real estate, rental/leasing, and several service composites.

13.6 Supporting U.S. quarterly product-tax exclusion is incomplete

The U.S. quarterly construction removes customs duties, federal excise taxes, and state/local sales taxes. It does not remove every product-tax subcomponent available in the annual decomposition. The annual data are richer; the quarterly data are better aligned with model frequency.

13.7 Supporting EA fiscal variables are not seasonally adjusted

The Eurostat government-finance extraction uses NSA data for `gov_10q_ggnfa`. The EA real GDP series used for the output gap is seasonally and calendar adjusted. This mismatch can introduce seasonal components into the fiscal regression. A future version should test seasonal adjustment, seasonal dummies, or four-quarter changes.

13.8 Output gaps are statistical, not structural

The output gap is the residual from a linear trend in log real GDP. It is not a model-implied output gap and not an official potential-output gap. This keeps the build transparent, but it is a simple proxy.

13.9 Debt concepts differ across countries and layers

The active sectoral U.S. panel inherits the U.S. debt concept from the Benchmark B panel, while the supporting aggregate quarterly layer uses federal debt as a percent of GDP for the United States and general-government gross debt for the euro area. These are not identical government-sector concepts. A fully harmonised build would use a common general-government debt concept for all regions.

13.10 Debt-feedback clipping is a model restriction

Zero entries in the runtime debt-feedback matrix are not necessarily estimated zeros. They are sign/closure restrictions imposed after unrestricted empirical sectoral rules generate negative debt feedbacks. This should be described as a model discipline, not as a data finding.

13.11 `Tbar_Y` remains model-implied

The steady-state production wedge remains:

```
Tbar_Y = 1/M - 1
```

This is a markup-offsetting object. It is not observed production-tax or subsidy data. The active database calibrates only dynamic production-wedge rules.

13.12 Labour and consumption fiscal blocks are not rebuilt here

The sectoral C2-SUT database is a production-wedge database. It does not solve the labour-tax wedge or consumption-tax wedge calibration. In the runtime, `Tbar_C`, `Tbar_N`, `Phi_tau_C_*`, and `Phi_tau_N_*` remain benchmarked or hand-set relative to the production-wedge data effort.

13.13 Generated artifacts are improved but still need governance

`calibration.jl` now imports the generated sectoral runtime file rather than only manually copying constants. This is an improvement. The next step is stronger artifact governance: checksums, generation timestamp in the runtime, and an automatic test that rejects mismatches between CSV coefficients and Julia matrices.

14 Recommended Referee-Facing Statement

A concise referee-facing description would be:

> The active production-wedge database is a sectoral C2-SUT calibration layer. For the United States, sectoral wedges are built from BEA Supply-Use tables by taking other taxes on production minus other subsidies on production minus subsidies on products, dividing by sector output at basic prices, and excluding the SUT row for taxes on products and imports, which includes import duties. Forty-four U.S. model-sector rules are estimated in no-intercept partial-adjustment form with output-gap and lagged-debt terms. EA, China, and ROW are not direct country-sector estimates; they scale the U.S. sectoral rules using the C2 country scaling layer. The unrestricted empirical debt-feedback coefficients are retained in the audit files, but the runtime clips negative production-wedge debt feedbacks to zero. The resulting runtime matrices are sectoral, but their non-U.S. rows are transferred calibrations rather than independently observed sectoral fiscal rules.

15 Reproducibility

From:

```
C:\CustomTools\MCMS_gvt\MCMS-private
```

rebuild the active sectoral dataset with:

```
python -B scripts\build_fiscal_rule_dataset_sectoral_c2_sut.py
```

The supporting aggregate Quarterly C2 dataset can be rebuilt with:

```
python -B scripts\build_fiscal_rule_dataset_quarterly_c2.py
```

verify the active Julia runtime integration by checking that `calibration.jl` imports and assigns the generated sectoral matrices:

```
Select-String -Path src\calibration.jl -Pattern "sectoral_c2_sut_runtime_matrices"
```

The active generated runtime file is:

```
input_data\fiscal_rules_sectoral_c2_sut\sectoral_c2_sut_runtime.jl
```

and the active Julia runtime imports it from:

```
src\calibration.jl
```